Paraneoplastic Cerebellar Degeneration (PCD): A Comprehensive Disease Characteristics Report

MONDO ID: MONDO:0044877 · **Category:** Complex (immune-mediated, non-Mendelian) **Report type:** Disease knowledge-base synthesis from literature (69 papers reviewed; 12 confirmed findings) **Evidence base:** Human clinical cohorts and case series, model-organism studies, in vitro immunology

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Summary

Paraneoplastic cerebellar degeneration (PCD) is a rare, immune-mediated, subacute pancerebellar syndrome that arises as a remote effect of cancer. It is **not** a genetic disease in the Mendelian sense; instead, the initiating lesion is a malignancy that **ectopically expresses a neuronal ("onconeural") antigen** normally restricted to cerebellar Purkinje cells. This aberrant expression breaks immune tolerance in genetically predisposed (HLA Class II) hosts, triggering an anti-tumor immune response that cross-reacts with the cerebellum. The dominant effector mechanism is **cytotoxic CD8+ T-cell-mediated destruction of Purkinje cells**, with autoantibodies serving primarily as diagnostic biomarkers rather than the direct cause of neuronal death (for the intracellular-antigen subtypes). The clinical result is subacute (weeks-to-months) progressive gait and limb ataxia, dysarthria, nystagmus, and dizziness, typically plateauing within ~6 months, often leaving patients severely and irreversibly disabled.

The single most important organizing principle to emerge from this investigation is the **antigen-location prognosis rule**: antibodies against *intracellular* antigens (anti-Yo/CDR2, anti-Hu/ANNA-1, Ma2) mark a T-cell-driven, largely irreversible process with poor neurological outcomes, whereas antibodies against *cell-surface* antigens (anti-DNER/Tr, anti-mGluR1) mark a more antibody-mediated, treatment-responsive process. Anti-Yo (PCA-1), directed against CDR2/CDR2L, is the most common variant, occurring almost exclusively in women with

gynecologic or breast cancer, and carries a dismal prognosis (≈84% of survivors unable to walk unassisted). Anti-Tr/DNER PCD, by contrast, occurs mostly in middle-aged men with Hodgkin lymphoma and is more treatable.

Management rests on two pillars applied urgently: **prompt tumor detection and treatment** (antibody-guided screening, with FDG-PET/CT central when conventional imaging is negative) and **immunotherapy** (corticosteroids, IVIG, plasma exchange, cyclophosphamide, rituximab). Good outcomes correlate with early diagnosis, low disability at presentation (mRS <3), absence of metastasis, and combined immunotherapy plus tumor-directed therapy. An emerging iatrogenic trigger is the class of **immune checkpoint inhibitors (ICIs)**, which can unmask or amplify pre-existing onconeural immunity. Population data indicate PCD is the second most common paraneoplastic neurological syndrome (≈28% of PNS), against an overall PNS incidence of ~0.89/100,000 person-years.

Section 1 — Disease Information

Overview. PCD is an autoimmune cerebellar syndrome triggered by an underlying (often occult) malignancy. It belongs to the broader family of paraneoplastic neurological syndromes (PNS) and immune-mediated cerebellar ataxias (IMCAs). The typical presentation is "the subacute development of pancerebellar deficits with a clinical plateau within 6 months" (PMID: 27606347).

Key identifiers. - **MONDO:** MONDO:0044877 - **MeSH:** Paraneoplastic Cerebellar Degeneration - **ICD-10:** G13.1 (Systemic atrophy primarily affecting central nervous system in neoplastic disease) / D48.9 with paraneoplastic manifestation - **Orphanet:** Paraneoplastic cerebellar degeneration (rare neurological PNS) - **OMIM:** Not applicable — PCD is not a heritable Mendelian disorder; no OMIM disease entry (relevant onconeural genes have Gene OMIM entries, e.g., *CDR2*).

Synonyms / alternative names. Paraneoplastic cerebellar degeneration; subacute cerebellar degeneration (paraneoplastic); anti-Yo/PCA-1 cerebellar ataxia (for that subtype); paraneoplastic cerebellar ataxia; onconeural cerebellar syndrome. Antibody-defined subtypes: PCA-1 (anti-Yo), PCA-Tr (anti-Tr/DNER), ANNA-1 (anti-Hu), ANNA-2 (anti-Ri), PCA-2 (anti-MAP1B).

Information source type. This report is derived from **aggregated disease-level resources** — systematic reviews, laboratory serology cohorts (Mayo Clinic, Barcelona), population-based epidemiology, and case series — rather than from individual EHR patient records.

Section 2 — Etiology

Primary cause. The causal factor is an **underlying malignancy** that ectopically expresses a neuronal antigen. In anti-Yo disease, "the Yo autoantibodies are directed against the Yo antigens, aberrantly overexpressed by tumor cells with frequent somatic mutations and gene amplifications" (PMID: 38494293). The disease is therefore fundamentally a **cancer-triggered autoimmune** process, not genetic, environmental (toxic), or infectious in origin.

Tumor-type risk factors (antibody-dependent). - Anti-Yo → gynecologic (ovarian, endometrial) and breast carcinoma; 96% female, 82% gynecologic cancer (PMID: 36334195). - Anti-Tr/DNER → Hodgkin lymphoma, middle-aged males (PMID: 34817790). - Anti-Hu, CV2/CRMP5, PCA-2/MAP1B, ZIC4 → small-cell lung cancer (SCLC), often in smokers. - Anti-Ma2 → testicular germ-cell tumors (young men <50), also lung/breast.

Genetic risk factors. Host susceptibility is conferred by **HLA Class II** haplotypes rather than causal coding mutations. High-resolution typing of 40 anti-Yo cases identified protective haplotypes (DPA1*01:03~DPB1*04:01, OR=0, p=0.0008) and an ovarian-cancer-specific susceptibility haplotype (DRB1*13:01~DQA1*01:03~DQB1*06:03, OR=5.4, p=0.0016), indicating "differential genetic susceptibility to anti-Yo per cancer and with a primary HLA Class II involvement" (PMID: 29306402).

Environmental / lifestyle risk factors. Tobacco smoking is a strong indirect risk factor via SCLC-associated subtypes (anti-Hu, CRMP5). No direct toxic, occupational, or infectious cause of PCD itself is established.

Iatrogenic trigger. Immune checkpoint inhibitors (anti-PD-1/PD-L1/CTLA-4) are an emerging cause; they can "amplify pre-existing onconeural immunity" (PMID: 42442848).

Protective factors. The only clearly documented protective factors are the HLA Class II protective haplotypes above. No dietary or lifestyle protective factors are established.

Gene–environment interaction. The model is: a tumor (environmental/somatic event) expressing an onconeural antigen, in a host carrying a permissive/susceptible HLA Class II genotype, produces tolerance breakdown. Molecular mimicry directs the specificity, and co-signaling molecules (checkpoint pathways) modulate the strength — a concept reinforced by the ICI-triggered cases (PMID: 39052041).

Section 3 — Phenotypes

The core phenotype is a **subacute, progressive pancerebellar syndrome** (HPO: HP:0001251 Ataxia; HP:0002070 Limb ataxia; HP:0002066 Gait ataxia).

Phenotype	HPO term	Type	Frequency / notes
Gait ataxia	HP:0002066	Clinical sign	Near-universal; often presenting feature
Limb ataxia / dysmetria	HP:0002070 / HP:0001310	Clinical sign	Very common
Dysarthria	HP:0001260	Clinical sign	Common
Nystagmus (incl. downbeat)	HP:0000639	Clinical sign	Common; downbeat characteristic
Vertigo / dizziness	HP:0002321	Symptom	Prodromal in ~two-thirds of anti-Yo patients (PMID: 36334195)
Diplopia / oscillopsia	HP:0000651 / HP:0011495	Symptom	Frequent
Truncal instability	HP:0002078	Clinical sign	Common
CSF lymphocytic pleocytosis	HP:0012310	Lab abnormality	Frequent, inflammatory CSF
Oligoclonal bands / elevated IgG index	HP:0032101	Lab abnormality	Frequent

Characteristics. Onset is **adult/late-adult** (median age ~60 for anti-Yo, ~68 in the population PNS cohort). Course is **subacute and progressive** over weeks to months, with "a clinical plateau within 6 months" (PMID: 27606347). Severity is typically **severe** for intracellular-antigen subtypes: despite treatment, 84% of anti-Yo survivors are unable to walk unassisted at follow-up (PMID: 36334195). "Vertigo and imbalance can be present early in the disease course in about two thirds of patients, as a prodromal phase" (PMID: 36334195).

Quality-of-life impact. Profound — most patients with intracellular-antigen PCD become wheelchair-dependent, non-ambulatory, and dependent for activities of daily living, with dysarthria impairing communication. Surface-antigen subtypes may recover substantially.

Extracerebellar features. Rare in isolated anti-Tr/DNER PCD (8%) (PMID: 34817790); anti-Hu and anti-Ma2 more often present with multifocal encephalomyelitis, limbic/brainstem involvement, and peripheral neuropathy.

Section 4 — Genetic / Molecular Information

PCD has **no causal germline disease gene**. The relevant molecules are the **onconeural target antigens** and the **host HLA Class II** susceptibility loci.

Onconeural antigens (target autoantigens).

Antibody (alias)	Target antigen / gene	Antigen location	Typical tumor
Anti-Yo (PCA-1)	CDR2 / CDR2L	Intracellular (cytoplasm)	Ovarian, breast
Anti-Tr (PCA-Tr)	DNER (Delta/Notch-like EGF-related receptor)	**Cell surface**	Hodgkin lymphoma
Anti-Hu (ANNA-1)	HuD / ELAVL family	Intracellular (nuclear)	SCLC
Anti-Ri (ANNA-2)	NOVA1/2	Intracellular (nuclear)	Breast, lung
Anti-Ma2	PNMA2 (Ma2)	Intracellular	Testicular germ-cell
Anti-CV2/CRMP5	CRMP5/DPYSL5	Intracellular	SCLC, thymoma
Anti-PCA-2	MAP1B	Intracellular	SCLC
Anti-mGluR1	GRM1 (metabotropic glutamate receptor 1)	**Cell surface**	Often non-paraneoplastic / lymphoma
Anti-Homer-3	HOMER3	Intracellular/postsynaptic	Breast adenocarcinoma
Anti-ZIC4	ZIC4	Intracellular	SCLC

The anti-Yo antigen designation and target are established: anti-Yo is "directed against cerebellar degeneration-related protein 2 (CDR2) and CDR2L" and is "the most common variant of paraneoplastic cerebellar degeneration" (PMID: 27606347). Anti-Tr binds "the extracellular domain of DNER" (PMID: 25745634).

Somatic vs germline. The genetic aberrations are **somatic in the tumor** — "aberrantly overexpressed by tumor cells with frequent somatic mutations and gene amplifications" of the Yo antigen genes (PMID: 38494293). There is no germline pathogenic variant driving PCD.

Modifier genes / functional consequences. HLA Class II haplotypes modify susceptibility per cancer type (PMID: 29306402). Epigenetic and chromosomal abnormality data specific to PCD are not established beyond tumor-level somatic changes.

HGNC / Gene identifiers: CDR2 (HGNC:1802), CDR2L (HGNC:14002), ELAVL4/HuD, PNMA2, DPYSL5/CRMP5, MAP1B, GRM1, DNER, HOMER3, ZIC4.

Section 5 — Environmental Information

- **Environmental/toxic factors:** No direct environmental toxin causes PCD. The relevant "environmental" exposure is **tobacco smoke**, acting indirectly by causing SCLC (the tumor substrate for anti-Hu, CRMP5, PCA-2, ZIC4 subtypes).
- **Lifestyle factors:** Smoking (via SCLC). No dietary triggers established.
- **Infectious agents:** None causative. PCD is a paraneoplastic (tumor-driven) autoimmune process, though molecular mimicry conceptually parallels post-infectious immune cerebellar ataxias in the broader IMCA family (PMID: 39052041).
- **Iatrogenic:** Immune checkpoint inhibitor therapy (see Sections 2, 6, 12).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. A **malignancy arises** (e.g., ovarian/breast carcinoma, SCLC, Hodgkin lymphoma) and, through somatic mutation/gene amplification, **ectopically expresses an onconeural antigen** (CDR2/CDR2L, HuD, DNER, Ma2) normally restricted to neurons → *results in* presentation of a neural self-antigen to the immune system in an immunogenic (tumor) context (PMID: 38494293).
2. In a **host carrying permissive HLA Class II haplotypes**, this ectopic antigen presentation *leads to* **breakdown of immune self-tolerance** (PMID: 29306402).
3. Tolerance breakdown *results in* an **anti-tumor adaptive immune response**: generation of onconeural autoantibodies AND antigen-specific T cells. (For surface antigens, the branch is antibody-dominant; for intracellular antigens, the branch is T-cell-dominant — see branch below.)
4. **Intracellular-antigen branch (Yo, Hu, Ma2):** cross-reactive **cytotoxic CD8+ T lymphocytes (CTLs)** traffic across the blood–brain barrier into the CSF/cerebellum → *leads to* antigen recognition of Purkinje cells → *results in* **CTL-mediated Purkinje cell apoptosis**/

death (PMID: 10632096; PMID: 9879687). Antibody alone is **insufficient** to cause degeneration (inferred from failed passive-transfer/immunization models) (PMID: 7707074; PMID: 7788981).

5. **Surface-antigen branch (DNER, mGluR1):** autoantibodies bind the extracellular domain of the target → *results in* receptor dysfunction/internalization that is potentially reversible → more treatment-responsive disease (PMID: 25745634; PMID: 41197574).
6. Purkinje cell loss *leads to* **loss of the sole output neuron of the cerebellar cortex** → *results in* **pancerebellar dysfunction** (gait/limb ataxia, dysarthria, nystagmus).
7. Progressive Purkinje depletion *leads to* **cerebellar atrophy** (radiographically visible late) and, for intracellular subtypes, **irreversible clinical disability** (PMID: 27606347).

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Tumor (ectopic onconeural antigen: CDR2/CDR2L, HuD, DNER, Ma2)
           │  (somatic mutation/amplification)
           ▼
HLA Class II–permissive host  ──►  Tolerance breakdown
           │
  ┌────────┴─────────┐
  ▼                  ▼
INTRACELLULAR Ag    SURFACE Ag
(Yo, Hu, Ma2)       (DNER, mGluR1)
  │                  │
CD8+ CTL response   Antibody binds
(dominant)          extracellular domain
  │                  │
Purkinje-cell        Receptor dysfunction
cytotoxic killing    (± reversible)
  │                  │
IRREVERSIBLE  ◄──── vs ────►  TREATMENT-RESPONSIVE
disability                    recovery
  │
  ▼
Purkinje cell loss → pancerebellar syndrome → cerebellar atrophy
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Molecular pathways / cellular processes. Adaptive immunity: MHC Class I/II antigen presentation, T-cell receptor engagement, CTL granule-mediated cytotoxicity, apoptosis (GO:0006915), and antigen-specific B-cell/plasma-cell antibody production. In active PCD CSF, ">75% of cells were CD3+ alphabeta T cells and 20–40% were activated T cells," and "activated cdr2-specific CTLs in the CSF contribute to Purkinje degeneration in PCD" (PMID: 10632096). CASPR2-associated cerebellar ataxia similarly shows combined CD8+ T-cell and CD138+ plasma-cell CSF infiltration (PMID: 22759321).

Protein dysfunction. For intracellular antigens the antigen is a normal neuronal protein (loss of Purkinje cells eliminates its function); for surface antigens (DNER, a Notch-pathway EGF-repeat receptor; mGluR1, essential for motor coordination/learning), antibody binding impairs signaling. Notably Homer-3's partner mGluR1A "is predominantly expressed in Purkinje cells where its function is essential for motor coordination and motor learning" (PMID: 35871640).

Immune system involvement. Central and defining — a cell-mediated (CTL) plus humoral autoimmune attack. Autoantibodies of multiple immunoglobulin classes bind Purkinje cytoplasm across species (PMID: 3346369).

Suggested GO / CL terms. GO:0006915 (apoptotic process), GO:0002456 (T cell mediated immunity), GO:0001913 (T cell mediated cytotoxicity), GO:0002376 (immune system process), GO:0019882 (antigen processing and presentation). **CL:** CL:0000121 (Purkinje cell — primary target), CL:0000794 (CD8-positive, alpha-beta cytotoxic T cell — effector), CL:0000786 (plasma cell), CL:0000909 (CD8-positive alpha-beta memory T cell).

Section 7 — Anatomical Structures Affected

- **Primary organ / system:** Central nervous system — the **cerebellum** (UBERON:0002037), specifically the cerebellar cortex (UBERON:0002129).
- **Primary cell target: Purkinje cells** (CL:0000121; UBERON: Purkinje cell layer UBERON:0002974). Purkinje-cell loss is the pathological hallmark; molecular-layer interneurons are also stained by autoantibodies (PMID: 3346369).
- **Secondary involvement:** Brainstem, limbic system, diencephalon, and peripheral nerves in overlap syndromes (anti-Hu encephalomyelitis, anti-Ma2 limbic/diencephalic/brainstem encephalitis, anti-Ri brainstem syndrome).
- **Tissue type:** Nervous tissue.
- **Subcellular compartments:** Depends on antigen — cytoplasm (CDR2/CDR2L; GO:0005737), nucleus (HuD, Ri; GO:0005634), plasma membrane / cell surface (DNER, mGluR1; GO:0005886), postsynaptic density (Homer-3; GO:0014069).
- **Localization / lateralization:** Bilateral, symmetric cerebellar involvement is typical (pancerebellar). FDG-PET may show bilateral cerebellar hypometabolism (early) or hyperperfusion in some inflammatory cases.

Section 8 — Temporal Development

- **Onset:** Adult to older-adult (median ~60–68 years). Onset pattern is **subacute** — evolving over weeks to a few months.
- **Prodrome:** Vertigo/imbalance precedes the full syndrome in ~two-thirds of anti-Yo patients (PMID: 36334195).
- **Progression:** Rapidly progressive during the active phase, then **plateaus within ~6 months** (PMID: 27606347), after which deficits are usually fixed (intracellular subtypes). Cerebellar atrophy appears late on MRI.
- **Course pattern:** Monophasic progressive-then-plateau for classic onconeural PCD; surface-antigen cases may relapse or remit with immunotherapy.
- **Duration:** Chronic/lifelong disability once the plateau is reached in intracellular-antigen disease.
- **Critical window:** There is a narrow therapeutic window **before irreversible Purkinje loss**; "early tumor detection, diagnosis, and PCD treatment are essential because any delay can result in the progression of the disorder and irreversible neurological damage" (PMID: 35501715).
- **Tumor timing:** The neurological syndrome usually **precedes** cancer diagnosis; screening should be repeated over time (tumors may surface years later) (PMID: 22157026).

Section 9 — Inheritance and Population

- **Epidemiology:** In a population-based 9-year Italian study (983,190 people), overall PNS incidence was **0.89/100,000 person-years** and prevalence **4.37/100,000**; "PNS developed in 1 in every 334 cancers" (PMID: 31552550). Cerebellar degeneration was the **second most common PNS at 28%** (after limbic encephalitis 31%): "Most common PNS were limbic encephalitis (31%), cerebellar degeneration (28%) and encephalomyelitis (20%)" (PMID: 31552550). Antibody specificities in that cohort: Yo 30%, Hu 26%, Ma2 22%; associated tumors lung 17%, breast 16%, lymphoma 12%.
- **Inheritance: Not heritable** — PCD is acquired/autoimmune. The only genetic contribution is HLA Class II susceptibility (polygenic/multifactorial predisposition), not a Mendelian inheritance pattern.

- **Penetrance / expressivity / anticipation / mosaicism / founder effects / consanguinity / carrier frequency:** Not applicable (no causal germline mutation).
- **Sex ratio:** Strongly subtype-dependent. Anti-Yo: 96% female (PMID: 36334195). Anti-Tr/DNER: predominantly middle-aged males (PMID: 34817790). Anti-Ma2: predominantly young males (testicular tumors). Overall PNS cohort ~52% female, median age 68 (PMID: 31552550).
- **Age distribution:** Adult/older adult, tracking the age of the underlying cancers.

Section 10 — Diagnostics

Clinical / CSF. CSF is frequently inflammatory: "lymphocytic pleocytosis, elevated protein, elevated IgG index, and oligoclonal bands" (PMID: 27606347; PMID: 35871640).

Antibody testing (the diagnostic cornerstone). Serum and CSF onconeural antibody panels: anti-Yo/CDR2/CDR2L, anti-Hu, anti-Ri, anti-Tr/DNER, anti-Ma2, anti-CV2/CRMP5, anti-PCA-2/MAP1B, anti-mGluR1, anti-Homer-3, anti-ZIC4. The anti-Tr/DNER recombinant cell-based assay reaches 100% sensitivity/specificity (PMID: 25745634). Rodent cerebellar tissue reliably reproduces the diagnostic Purkinje-cytoplasm immunostaining (PMID: 3346369).

Imaging. "Magnetic resonance imaging of the brain is often normal in the early stages, with cerebellar atrophy seen later" (PMID: 27606347). FDG-PET can show cerebellar hypometabolism.

Tumor screening (antibody-guided, time-critical). Per the EFNS task force, "the nature of antibody, and to a lesser extent the clinical syndrome, determines the risk and type of an underlying malignancy," and "for screening of the thoracic region, a CT-thorax is recommended, which if negative is followed by fluorodeoxyglucose-positron emission tomography (FDG-PET)" (PMID: 20880069). FDG-PET/CT detected malignancy in ~19% of suspected PNS patients and is recommended regardless of antibody status (PMID: 22157026). Screening should be **repeated** if initially negative, because the tumor usually postdates neurological onset.

Diagnostic criteria. The 2021 updated PNS diagnostic criteria replaced "classical syndromes" with "high-risk phenotypes," reclassified antibodies as high-risk (>70% cancer association) vs intermediate-risk (30–70%), and introduced the **PNS-Care Score** combining phenotype, antibody, cancer presence, and follow-up to grade diagnoses as definite/probable/possible (PMID: 34006622).

Differential diagnosis. Hereditary/degenerative ataxias, multiple system atrophy (cerebellar type), toxic/metabolic cerebellar injury, gluten ataxia, anti-GAD ataxia, post-infectious cerebellitis, primary autoimmune cerebellar ataxia (PACA), and metastatic/leptomeningeal disease (PMID: 39052041; PMID: 35618871).

Section 11 — Outcome / Prognosis

Antigen-location prognosis rule (central finding). Intracellular-antigen antibodies (Yo, Hu) predict irreversible disability; surface-antigen antibodies (DNER, mGluR1) predict a treatable course. In anti-Yo disease, despite treatment, **84% of survivors are unable to walk unassisted** at follow-up (PMID: 36334195). In anti-mGluR1 cerebellar syndrome, "the majority of patients showed clinical improvement (n = 31)" of 42 (PMID: 41197574).

Predictors of good outcome. In a 97-patient PNS cohort with good outcome in 54.6%, "factors associated with good outcome were: early diagnosis, mRS <3 at presentation, absence of metastatic disease, and adjuvant immunotherapy" (PMID: 33911377).

Oncologic paradox. PNS can prompt earlier cancer detection, conferring a survival advantage despite neurological morbidity: ANNA1/Hu-IgG PNS + SCLC patients had "a 41% lower hazard of death" than SCLC-only patients (HR=0.59, 95% CI 0.37–0.96) (PMID: 42233990).

Subtype-specific prognosis. Anti-Ma2-only patients fare significantly better than anti-Ma (Ma1+Ma2) patients — "the clinical outcome was significantly better in the anti-Ma2 group" (PMID: 27460184). ICI-related cerebellar ataxia: 46% improved but with residual disability (PMID: 39153058). In ICI-related PNS more broadly, risk-antibody positivity carried 29% mortality vs 10% in antibody-negative patients (P=0.012) (PMID: 41488641).

Morbidity / QoL. High disability burden: wheelchair dependence, dysarthria, and loss of independence dominate the intracellular-antigen subtypes.

Section 12 — Treatment

Treatment rests on **two urgent pillars**: (1) prompt tumor removal/therapy and (2) immunotherapy.

Tumor-directed therapy. Surgery, chemotherapy, or radiotherapy of the underlying malignancy is the mainstay and can stabilize or improve neurological symptoms.

Immunotherapy (NCIT: Immunotherapy). - Corticosteroids / IV methylprednisolone pulses (NCIT: Methylprednisolone) - Intravenous immunoglobulin (NCIT: Intravenous Immunoglobulin Therapy) - Plasma exchange (NCIT: Plasmapheresis) — can give repeated benefit in some cases (PMID: 31142706) - Cyclophosphamide (NCIT: Cyclophosphamide) - Rituximab / anti-CD20 (NCIT: Rituximab); ofatumumab reported in refractory anti-Yo (PMID: 39737186)

Response by subtype. "Patients with surface receptor autoimmunity … usually show a good response to treatment," in contrast to classical (intracellular) onconeural PNS (PMID: 29327271). Anti-Yo/intracellular cases often deteriorate despite aggressive therapy (PMID: 35501715). Early rituximab benefited non-tumor anti-DNER (PMID: 37991702) and anti-mGluR1 (PMID: 40760473) cases.

ICI-related PCD management. Discontinue the checkpoint inhibitor and start immunosuppression; outcomes are variable, and rechallenge can provoke relapse (PMID: 37151179).

Supportive / rehabilitative. Symptomatic agents (e.g., 4-aminopyridine for downbeat nystagmus/oscillopsia) plus physical, occupational, and speech therapy.

Overall. Good outcome in ~55% when combined immunotherapy + tumor treatment is applied early (PMID: 33911377).

Section 13 — Prevention

- **Primary prevention:** Reducing cancer risk (e.g., smoking cessation to reduce SCLC-associated subtypes) is the only meaningful primary preventive lever; there is no vaccine or specific prophylaxis.
- **Secondary prevention (most actionable):** Early recognition of a subacute cerebellar syndrome, prompt antibody testing, and **antibody-guided tumor screening with FDG-PET** to detect and treat the malignancy before irreversible Purkinje loss (PMID: 20880069; PMID: 22157026). In known cancer survivors, PCD can herald relapse (e.g., Hodgkin lymphoma) — vigilance enables early re-treatment.

- **Tertiary prevention:** Immunotherapy plus rehabilitation to limit disability once disease is established.
- **Pre-ICI risk stratification:** Pre-treatment screening for onconeural antibodies before checkpoint-inhibitor therapy may identify high-risk patients (PMID: 41488641).
- **Genetic counseling / carrier screening / immunization:** Not applicable (non-heritable, non-infectious).

Section 14 — Other Species / Natural Disease

- **Taxonomy:** Naturally occurring PCD is essentially a **human (Homo sapiens, NCBI:txid9606) disease**. No well-characterized spontaneous paraneoplastic cerebellar degeneration is documented in companion animals as a defined entity.
- **Related veterinary autoimmune cerebellar/encephalitic disease:** Neuronal-autoantibody encephalitis is increasingly recognized in cats — including a reported case of anti-mGluR1 antibodies in a cat (PMID: 42656301) — but this was not clearly paraneoplastic.
- **Orthologous genes:** CDR2/CDR2L, ELAVL4/HuD, Grm1, Dner, Pnma2, Homer3 have rodent orthologs (mouse *Cdr2*, *Grm1*, *Dner*), enabling model work.
- **Comparative pathology:** Autoantibodies from human PCD sera bind Purkinje-cell cytoplasm in human, rat, and mouse cerebellum, indicating cross-species antigen conservation (PMID: 3346369).
- **Zoonotic potential:** None (autoimmune, non-transmissible).

Section 15 — Model Organisms

Attempts to build a faithful animal model have repeatedly **failed to reproduce degeneration by antibody alone**, which is itself the key mechanistic evidence for T-cell causation.

Model approach	Result	Interpretation
Active immunization of mice (BALB/c, C3H, C57BL/6, SJL/J) with recombinant Yo	"All the strains produced high anti-Yo antibody titer but none developed cerebellar ataxia or showed Purkinje cell loss" (PMID: 7788981)	Antibody insufficient
Passive transfer of PCD patient IgG ± complement/macrophages into rodent brain	IgG taken up by Purkinje cells >36 h without cell loss; "could not be the sole cause of Purkinje cell loss" (PMID: 7707074, PMID: 7788964)	Antibody insufficient
DNA immunization against pcd17/cdr2	Induced antibodies AND CTLs, but "neither clinical nor pathological changes consistent with significant cerebellar degeneration" (PMID: 11771954)	Immunity inducible; degeneration not recapitulated — model limitation
In vitro human CTLs (HLA-A24) + recombinant Yo on autologous dendritic cells	CTLs reacted with Yo; "cytotoxic T cells are involved in Purkinje cell loss in PCD" (PMID: 9879687)	Positive evidence for CTL effector

Model type: Mammalian (mouse, rat, SCID mouse) and in vitro human cellular immunology. **Genetic models:** immunization/transgene (DNA immunization) rather than knockout disease models. **Phenotype recapitulation:** Poor — antibodies and even antigen-specific CTLs can be generated without overt cerebellar degeneration, a major limitation likely reflecting incomplete CNS T-cell trafficking or additional required signals. **Applications:** Established that antibody is not the sole effector and pointed to CTLs; useful for diagnostic reagent validation (rodent cerebellum for immunostaining).

Mechanistic Model / Interpretation

PCD is best understood as **collateral autoimmune damage from an anti-tumor immune response**. The unifying model is antigenic: a tumor ectopically expresses a neuronal protein → tolerance breaks in an HLA-Class-II-permissive host → adaptive immunity attacks both tumor and cerebellum. The **antigen's subcellular location dictates the effector mechanism and therefore the prognosis**:

- **Intracellular antigens** (CDR2/CDR2L, HuD, Ma2) can only be seen by the immune system as MHC-presented peptides → **CD8+ CTL cytotoxicity** → Purkinje-cell death that is rapid, irreversible, and poorly antibody-treatable. The antibodies are bystander biomarkers.
- **Surface antigens** (DNER, mGluR1) are directly accessible to antibodies → **antibody-mediated receptor dysfunction** that is often reversible → treatment-responsive disease.

This single axis explains the epidemiology (tumor associations by antibody), the diagnostics (antibody-guided screening), the treatment response gradient, and the prognosis. The failed animal models are not a gap but positive evidence: they demonstrate that circulating antibody, complement, and macrophages cannot kill Purkinje cells, forcing the conclusion that the effector is the cytotoxic T cell.

Evidence Base (key literature)

PMID	Contribution	Type
27606347	Anti-Yo most common PCD variant; CDR2/CDR2L; subacute pancerebellar course, MRI evolution	Human review
36334195	379-patient anti-Yo systematic review: 96% female, 82% gynecologic, 84% non-ambulatory	Human systematic review
38494293	Ectopic tumor overexpression of Yo antigens with somatic mutations/amplifications	Human review
34817790	85-patient anti-Tr/DNER review: middle-aged males, 91% tumor, Hodgkin lymphoma	Human systematic review
25745634	Anti-Tr binds extracellular DNER; 100% sensitive/specific CBA	Human/in vitro
10632096	Activated cdr2-specific CTLs in CSF drive degeneration	Human immunology
9879687	Patient CTLs react with recombinant Yo — CTL effector role	In vitro
11771954	DNA immunization induces antibody+CTL but no degeneration	Mouse model
7788981 / 7707074 / 7788964	Antibody alone insufficient for Purkinje loss	Mouse/rat model
29306402	HLA Class II susceptibility/protective haplotypes in anti-Yo	Human genetics
31552550	Population-based PNS incidence 0.89/100k; PCD = 28% of PNS	Human epidemiology
42233990	Anti-Hu survival paradox (41% lower death hazard w/ SCLC)	Human cohort
41197574	Anti-mGluR1: majority improve with immunotherapy	Human systematic review
33911377	Predictors of good outcome; 54.6% good outcome	Human cohort
20880069 / 22157026	Antibody-guided FDG-PET tumor screening	Guideline/human

PMID	Contribution	Type
34006622	2021 PNS diagnostic criteria & PNS-Care Score	Consensus guideline
27460184	Anti-Ma2 testicular tumors 40%; Ma2-only better outcome	Human series
39153058 / 42442848	ICI-related cerebellar ataxia; amplification of onconeural immunity	Human cohort/case
3346369	Cross-species Purkinje-cytoplasm antibody binding; diagnostic substrate	Human/animal

Limitations and Knowledge Gaps

1. **No faithful animal model** of the degeneration exists; the CTL effector mechanism is inferred from failed antibody-transfer experiments plus in vitro human CTL reactivity, not from a reproducible in vivo lesion.
2. **Rarity and heterogeneity** limit prospective, controlled treatment data; most evidence is from case series, retrospective cohorts, and systematic reviews of case reports — susceptible to publication and referral bias (serology-lab cohorts over-represent antibody-positive cases).
3. **Molecular detail of tolerance breakdown** (why specific tumors over-express onconeural antigens, and the exact epitope-spreading/mimicry events) is incompletely defined.
4. **Prognostic biomarkers** beyond antibody class/antigen location are lacking; no validated molecular predictor of immunotherapy response.
5. **Epigenetic, transcriptomic, proteomic, and single-cell profiling** of PCD cerebellum are largely absent from the literature reviewed — a genuine data gap.
6. **HLA association data** derive from a single modest cohort (n=40); replication across ancestries is needed.

Proposed Follow-up Experiments / Actions

1. **Single-cell / spatial profiling** of PCD-affected cerebellum and matched CSF (CITE-seq, TCR sequencing) to define the clonality and antigen-specificity of infiltrating CD8+ T cells and confirm the CTL model in situ.
2. **Humanized HLA-transgenic mouse models** expressing onconeural antigens in a tumor context, with adoptive transfer of antigen-specific CD8+ T cells, to finally recapitulate Purkinje-cell loss.
3. **Multi-center prospective registry** with standardized PNS-Care scoring, antibody subtyping, and mRS trajectories to quantify outcomes and treatment effects by antigen location.
4. **Replication of HLA Class II associations** in larger, ancestrally diverse cohorts, extended to non-Yo subtypes.
5. **Trials of T-cell-directed immunotherapy** (e.g., agents targeting CD8+ CTLs or trafficking) for intracellular-antigen PCD, where B-cell-directed therapy underperforms.
6. **Pre-ICI onconeural antibody screening protocols** with prospective evaluation of whether screening plus surveillance reduces severe checkpoint-inhibitor cerebellar toxicity.
7. **Biomarker discovery** (CSF proteomics/neurofilament light) to identify early, treatable-window markers before irreversible atrophy.

Report compiled from 69 reviewed publications and 12 confirmed findings across 5 investigative iterations. Evidence types are labeled per claim; PMIDs link to primary sources.
